

SOFTWARE ENGINEERING

AI Travel Planning Assistant

Travito

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Table of Contents

- 1. Introduction
 - 1.1 Purpose
 - 1.2 Scope
 - 1.3 Intended Audience
 - 1.4 Document Conventions
- 2. Overall Description
 - 2.1 Product Perspective
 - 2.2 Product Functions
 - 2.3 User Characteristics
 - 2.4 Operating Environment
 - 2.5 Design Constraints
 - 2.6 Assumptions and Dependencies
- 3. Functional Requirements
- 4. Non-Functional Requirements
- 5. System Architecture
- 6. Database Requirements
- 7. Future Enhancements

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the AI Travel Planning Assistant (Travito). The application assists travelers by generating AI-powered travel itineraries, recommending accommodations, managing travel expenses, preparing packing checklists, and answering destination-related questions using Artificial Intelligence.

1.2 Scope

Travito is a web-based travel planning platform that combines Artificial Intelligence with traditional travel management features. The system enables users to:

- Create secure user accounts
- Generate personalized travel plans
- Receive hotel recommendations
- Track travel expenses
- Maintain packing checklists
- Interact with an AI travel assistant
- Store previous trips for future reference

1.3 Intended Audience

This document is intended for:

- Project Supervisor
- Academic Evaluation Committee
- Software Developers
- Testers
- Future Maintenance Team

1.4 Document Conventions

The document follows the IEEE Software Requirements Specification standard.

- Bold text represents important terms.
- Code Font represents APIs, database tables, and system components.
- FR denotes Functional Requirement.
- NFR denotes Non-Functional Requirement.

2. Overall Description

2.1 Product Perspective

Travito follows a Client-Server architecture. The frontend communicates with the backend through REST APIs. The backend stores user information in SQLite and communicates with the Groq AI service for intelligent travel planning.

Layer	Technology
Frontend	React.js
API Layer	REST APIs (Express.js)
Backend	Node.js
Database	SQLite
AI Service	Groq AI API (Llama 3.3)

2.2 Product Functions

Major system functions include:

- User Registration and Login
- AI Travel Itinerary Generation
- Hotel Recommendations
- Budget Management
- Expense Tracking
- Packing Checklist
- AI Travel Chat Assistant
- Saved Trips Management

2.3 User Characteristics

The intended users include solo travelers, families, couples, friends, students, and tourists. Users are expected to possess basic computer and internet usage skills.

2.4 Operating Environment

Category	Technology / Platform
Frontend	React.js, HTML5, CSS3, JavaScript
Backend	Node.js, Express.js
Database	SQLite
AI Service	Groq API, Llama 3.3 Model
Operating Systems	Windows, Linux, macOS
Browsers	Google Chrome, Microsoft Edge, Mozilla Firefox

2.5 Design Constraints

- Internet connection required for AI features.
- Groq API usage depends on available API quota.
- SQLite supports a single local database.
- JWT is used for authentication.

2.6 Assumptions and Dependencies

- Users have stable internet connectivity.
- Groq API remains available.
- Browser supports JavaScript.
- Server environment supports Node.js.

3. Functional Requirements

3.1 User Authentication Module

Requirement ID	Description
FR-1.1	The system shall allow users to register using name, email, and password.
FR-1.2	Passwords shall be hashed using BCrypt before storage.
FR-1.3	The system shall authenticate users and generate JWT tokens valid for seven days.
FR-1.4	Unauthorized users shall not access protected resources.

3.2 AI Travel Planner Module

Requirement ID	Description
FR-2.1	Users shall enter destination, travel dates, budget, travel style, and companion type.
FR-2.2	The system shall request itinerary generation from the Groq AI API.
FR-2.3	If the AI service is unavailable, the system shall provide predefined travel templates.
FR-2.4	Users shall preview and save generated itineraries.

3.3 Hotel Recommendation Module

Requirement ID	Description
FR-3.1	The system shall recommend hotels according to the user's budget.
FR-3.2	Hotels shall be categorized as Budget, Mid-range, and Luxury.
FR-3.3	Recommendations shall be generated together with the itinerary.

3.4 Expense Tracker Module

Requirement ID	Description
FR-4.1	Users shall add travel expenses.
FR-4.2	Users shall edit or delete expenses.
FR-4.3	The system shall calculate total spending automatically.
FR-4.4	Budget utilization shall be displayed through charts.

3.5 Packing Checklist Module

Requirement ID	Description
FR-5.1	The system shall categorize packing items into Documents, Clothing, Electronics, Toiletries, and Others.
FR-5.2	Users shall mark items as packed or unpacked.
FR-5.3	Packing status shall be stored permanently.

3.6 AI Chat Assistant Module

Requirement ID	Description
FR-6.1	Users shall communicate with the AI assistant.
FR-6.2	The assistant shall answer destination-specific travel questions.
FR-6.3	The current trip information shall be provided as AI context.
FR-6.4	Chat history shall be maintained during the user session.

4. Non-Functional Requirements

4.1 Performance

Requirement ID	Description
NFR-1	Page loading time shall remain below 150 milliseconds.
NFR-2	AI responses shall be generated within approximately 3.5 seconds.

4.2 Security

Requirement ID	Description
NFR-3	Passwords shall be encrypted using BCrypt hashing.
NFR-4	JWT authentication shall secure protected APIs.
NFR-5	User sessions shall expire automatically after seven days.

4.3 Reliability

Requirement ID	Description
NFR-6	The database shall automatically initialize if not found.
NFR-7	The application shall recover gracefully from database file locks.
NFR-8	The system shall continue functioning using mock data when AI services are unavailable.

4.4 Usability

Requirement ID	Description
NFR-9	The user interface shall be responsive across desktop, tablet, and mobile devices.
NFR-10	The system shall support screen widths from 320 pixels onward.

5. System Architecture

The application consists of four primary components: React Frontend, Express Backend, SQLite Database, and Groq AI Service. The frontend communicates with the backend using REST APIs. The backend processes user requests, stores data in SQLite, and interacts with the Groq AI service to generate intelligent travel recommendations.

Component	Technology	Role
React Frontend	React.js, HTML5, CSS3	Renders UI and communicates with backend via REST APIs
Express Backend	Node.js, Express.js	Handles API requests and business logic
SQLite Database	SQLite	Persists user data, trips, expenses, chats
Groq AI Service	Groq API, Llama 3.3	Generates intelligent travel itineraries and responses

6. Database Requirements

The system uses an SQLite database with the following main tables:

Users	Stores user account information (name, email, hashed password)
Trips	Stores travel itineraries generated by the AI planner
Expenses	Stores individual expense records for each trip
PackingItems	Stores packing checklist items and their packed status
ChatHistory	Stores AI conversation history per user session

7. Future Enhancements

Future versions of Travito may include the following enhancements:

- Flight Booking Integration
- Google Maps Navigation
- Weather Forecasting
- Currency Conversion
- Multi-language Support
- Offline Trip Mode
- AI Image-Based Destination Suggestions
- Push Notifications

8. Conclusion

This Software Requirements Specification provides a comprehensive description of the functional and non-functional requirements for the AI Travel Planning Assistant (Travito). It serves as the primary reference for system development, implementation, testing, deployment, and future maintenance while ensuring that the software meets both user needs and academic project requirements.